# Three-dimensional Blood Vessel Detection from Small Number of Views CT Imaging

Esraa A. Mohamed and Essam A. Rashed

***Abstract*—The identification of blood vessels is an important image processing technique in several medical imaging applications. Detection of blood vessels in computed tomography (CT) images is essential in diagnosis and surgical planning. Computed tomography (CT) is a common imaging modality used for blood vessels imaging. It can provide a high quality images for both diagnosis and treatment purposes. However, it is known that CT is a source of high radiation dose especially for cases required a follow up scans in short time. Reducing the dose is knows to produce statistical noise and artifacts that significantly reduce the image quality. In this paper, we propose a method for three-dimensional (3D) blood vessel detection from CT images reconstructed from small number of projection views. The proposed method is implemented in two stages. First, is a preprocessing stage to reduce noise. In the second stage, we use skeletonization method to extract the centerline of 3D object. Then, we use the blood vessels skeleton to construct a graph model for 3D vascular network. Finally, we use the graph model to detect blood vessels. The proposed method is tested using 3D blood vessel-like phantom and good results are achieved.**

## I. Introduction

Accurate and efficient detection of blood vessels in medical images is important in several applications such as accurate diagnosis and surgical planning. Computed Tomography (CT) is a common imaging modality used for blood vessels imaging [1]. It can provide a high quality images for both diagnosis and treatment purposes. However, CT is known as a source of high radiation dose especially for cases required a follow up scans in short time. Reducing the number of projection views in CT imaging is known to produce statistical noise and artifacts that significantly reduce the image quality. Figure 1 illustrates difference between using large number of projection views and small number of projection views. So, the detection of blood vessels from CT images reconstructed from small number of projection views is challenging due to noise, small artifacts and low-resolution of vessel structures.

In this paper, we propose a method for three-dimensional (3D) blood vessel detection from CT images reconstructed from small number of projection views. The proposed method is implemented in two stages. First, is a preprocessing stage. In this stage, median filter is used reduce noise. In addition to Otsu's thresholding method [2], which is used to convert the filtered image to binary format. In the second stage, we use skeletonization method to extract the centerline of 3D tubular object. Then, we use the blood vessels skeleton to construct a graph model for 3D vascular network. Finally, we use the graph model to detect blood vessels. The proposed method is tested using 3D blood vessel-like phantom and satisfactory results are achieved.

E.A.Mohamed and E.A.Rashed are with the Image Science Lab., Dept. of Mathematics, Faculty Science, Suez Canal University, Ismailia 41522, Egypt (E-mail: esraa_ahmed9033@yahoo.com).

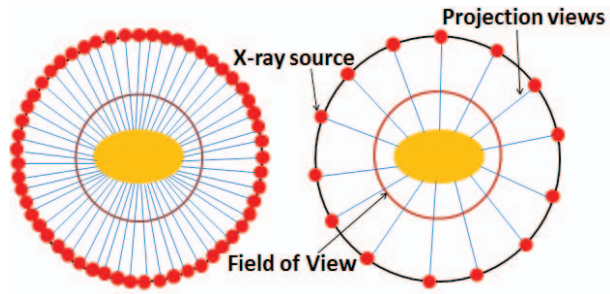


Fig. 1: Sketch representing sample for large number of pro-jection views and small number of projection views.

The structure of the paper is as following. In Section II the proposed method is described. Experimental results are provided in Section III. Finally, Section IV is for the conclusion.

## II. Methods

### A. Image preprocessing

Consider that we have CT image $I$ reconstructed from small number of projection views. First, we use median filter by Eq. (1).

$$G(x,y,z) = \text{Median}(I(i,j,k))\ \forall i \in \Omega_x, j \in \Omega_y, k \in \Omega_z, \tag{1}$$

where $\Omega_x$, $\Omega_y$ and $\Omega_z$ are a neigbourhood intervals around $x$, $y$, and $z$ respectively. Median filter is used to reduce the statistical noise and low frequency artifacts. Then, the filtered image $G$ is converted to binary image using Otsu's thresholding method [2]. Otsu's thresholding method is aimed to find the optimal value for the global threshold based on the interclass variance maximization. We can calculate the interclass variance as in Eq. (2)

$$\sigma^2_{interclass}(T) = n_B(T) n_O(T) \left[\mu_B(T) - \mu_O(T)\right]^2 \tag{2}$$

where $0 < T < L-1$, $n_B(T) = \sum_{i=0}^{T-1} G(i)$, $n_O(T) = \sum_{i=T}^{L-1} G(i)$, $\mu_B(T)$ and $\mu_O(T)$ are the mean of the background pixels (below threshold) and the foreground pixels (above threshold) respectively and $[0, L-1]$ is the range of intensity level. The optimal thresholding value, $T^*$, satisfies Eq. (3)

$$\sigma^2_{interclass}(T^*) = \text{Max}_{0<T<L-1} \sigma^2_{interclass}(T) \tag{3}$$

978-1-5090-1642-6/16/$31.00 ©2016 IEEE

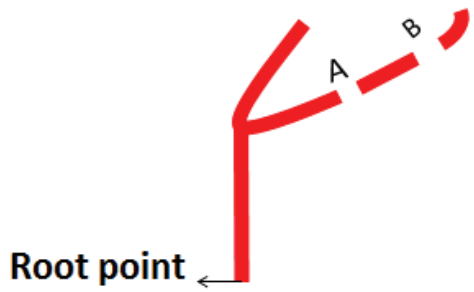


Fig. 2: Demonstration of simple vessels skeleton with gaps. The tree construction is robust w.r.t. gaps **A**, **B** between the root point and endpoints. Gap **A**: represent two endpoints that one of them is connected to the root and the other is not connected. Gap **B**: represent two endpoints that are not connected to the root point.

### B. Skeletonization

Skeletonization aims to extract a region-based shape feature representing the general form of an object. We have applied skeletonization to extract the centerline of a 3D tubular object. We used binary thinning algorithm [3] to find the centerlines (skeleton) of the 3D object. The main idea is to erode the objects surface iteratively until only the skeleton remains. Erosion [5] has to be performed symmetrically in order to guarantee medial position of the skeleton lines such that the connectedness of the object is preserved.

A number of tests are performed in order to check if a voxel can be eroded from the object. This is done for all voxels in the image and repeated until no more change occurs. The current voxel can be deleted if:

1) the current voxel is a surface voxel in order to guarantee that the centerlines are not shifted to one side of the object.
2) the current voxel is not the end of a line, because the centerlines are to remain.
3) the deletion of the current voxel would not change the Euler characteristic [4], *i.e.* if no holes are created when deleting this voxel.
4) the current voxel is a simple point, that is the deletion does not change the number of connected objects.

All these tests are performed in parallel to all voxels of the 3D image. Finally, a sequential re-checking method is performed to ensure that connectivity is preserved even after deletion of neighbor voxels.

### C. Tree construction

We consider every voxel belonging to the skeleton as tree node and this node is connected by edge to its $((n+2)^3-1)$-neighbors where, $n = 1, 3, 5, \cdots$. This tree has a root which is determined manually and the branches of the tree follow the skeleton. Then, we determine the endpoints of this tree. The aim of the proposed method is to detect the path from root point to endpoints but we can not use the skeleton directly because it may contain gaps.

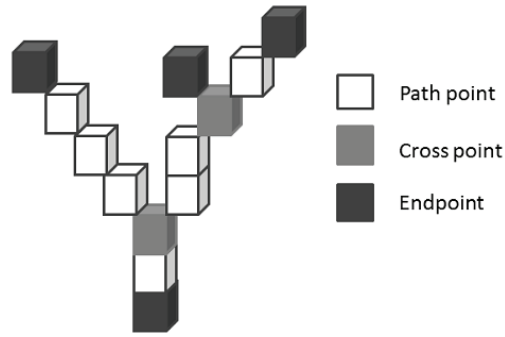


Fig. 3: Example for tree voxels identification.

*1) Gaps:* The skeleton may contain gaps between the root and endpoints (see Fig. 2). In order to close remaining gaps, we look for Two endpoints that one of them or the two are not connected to the root, calculate the distance between them and fill the gap between the detected nodes if the distance between them is less than thresholding value.

*2) Identification of tree voxels:* We classify voxels which belongs to the tree into three types of points (see Fig. 3):

- End point: which have only one voxel neighborhood.
- Path point: which have exactly two voxel neighborhood.
- Crossing point: which have more than two voxel neighborhood.

Equation (4) is used for counting number of neighbors *Ni* for each voxels belonging to the tree, where *S* is the skeleton image after filling the gaps, *S(x,y,z)* is non-zero voxel.

$$Ni(S(x,y,z)) = \text{Count}\,\{(S(i,j,k))\} - 1 \;\forall i \in \Omega_x, j \in \Omega_y, k \in \Omega_z \tag{4}$$

where $\Omega_x$, $\Omega_y$ and $\Omega_z$ are a neigbourhood intervals around $x$, $y$, and $z$ respectively.

### D. Vessel detection

After constructing the graph model and determining the root point and the end points, we use depth-first algorithm [6] to detect the path from root to each endpoint. The path from the root to end point represent blood vessel. Figure 4 represents flowchart of the proposed method.

## III. Experimental Results

A computer simulation study was implemented using FORBILD lung phantom [7] shown in Fig. 5(a). The phantom size is $256^3$ pixels. Proposed method is performed in two stages. The first stage is preprocessing and is shown in Fig. 5. Projection data is measured over 25 projection views. Reconstruction is performed using Algebric Reconstruction Technique (ART) [8]. Results of reconstruction is shown in Fig. 5(b). We note that streaks and low-frequency artifacts are appeared in the reconstructed image due to the reduction of projection views. Then, we use median filter to reduce noise. The filtered image is shown in Fig. 5(c). The filtered image is converting into binary in Fig. 5(d). The second stage is detection of blood vessel. Steps for second stage is shown in Fig. 6. Extracting the skeleton of the phantom is shown

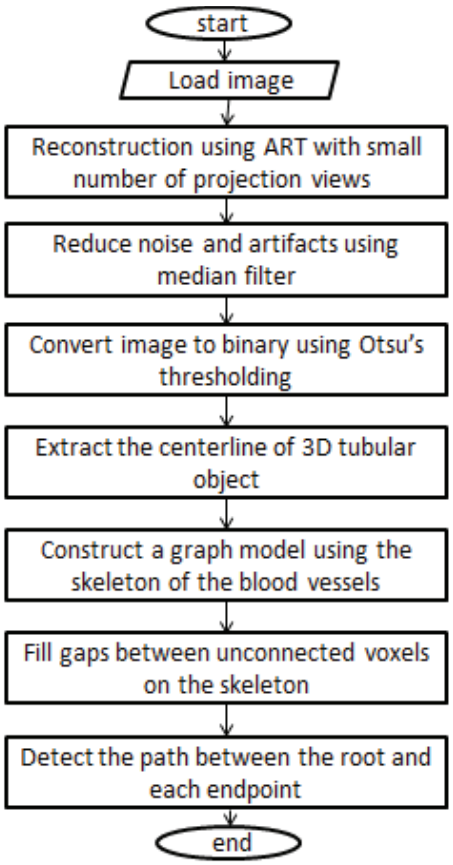


Fig. 4: Flowchart of the proposed method.

TABLE I: Root mean square error between skeleton of original image pixels and blood vessel detection image pixels using proposed method with different number of views.

| Number of views | RMSE |
|---|---|
| 20 | $3.17 \times 10^{-4}$ |
| 25 | $3.13 \times 10^{-4}$ |
| 35 | $3.73 \times 10^{-4}$ |
| 50 | $3.64 \times 10^{-4}$ |

in Fig. 6(a). Detection of gaps between skeleton voxels is shown in Fig. 6(b). The skeleton after filling gaps between unconnected voxels is shown in Fig. 6(c). Then, path between the root point and endpoints are detected as shown in Fig. 6(d). We use the root mean square error $(RMSE)$ as image quality metric.

$$RMSE = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\left(f_j - \hat{f}_j\right.^{2}} \tag{5}$$

where $f_j$ represents blood vessel detection image pixels and $\hat{f}_j$ represents the skeleton of original image pixels.

Table I represents calculation of RMSE with different number of projection views. Calculation of RMSE with different thresholding values is shown in table II. From table I and table II, we can note that the number of projection views and thresholding value which we use in the experimental results have the least RMSE values.

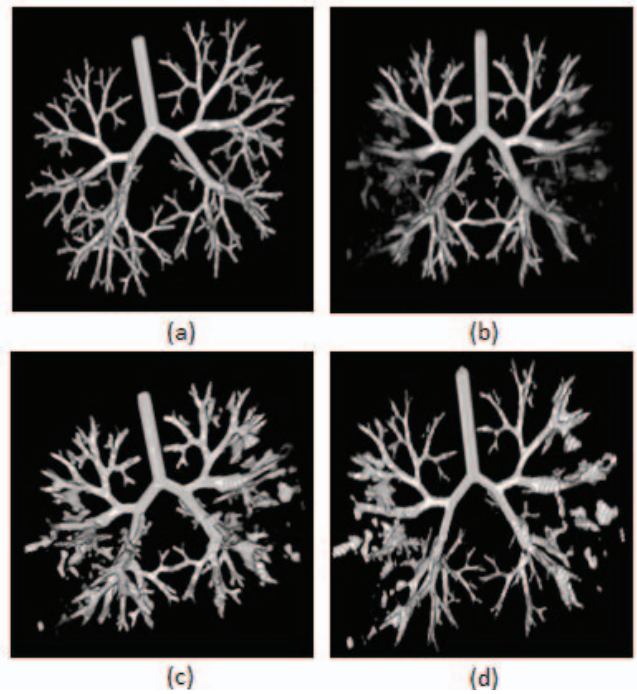


Fig. 5: Preprocessing step. (a) Original image. (b) Recon-structed image. (c) Filtered image. (d) Binary image.

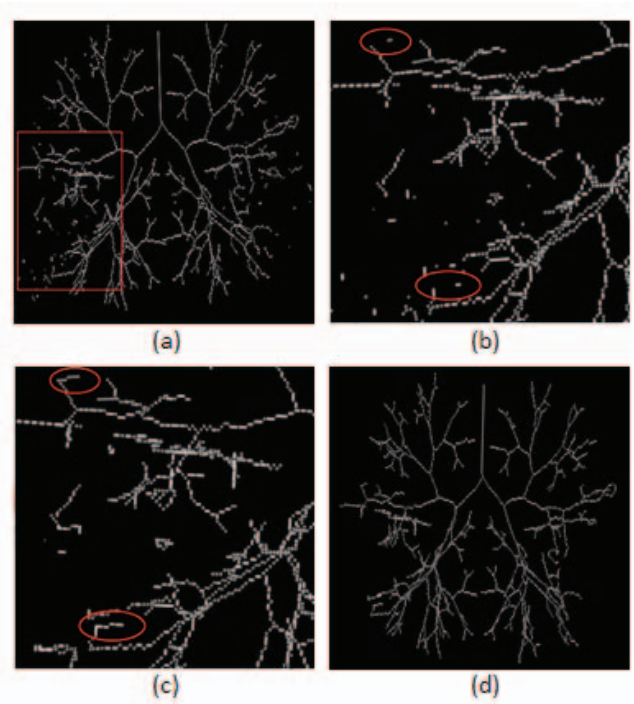


Fig. 6: Steps for blood vessel detection. (a) Skeleton of the binary image. (b) Detection of skeleton gaps (c) Filling gaps between skeleton voxels. (d) Blood vessel detection.

TABLE II: Root mean square error between skeleton of original image pixels and blood vessel detection image pixels using proposed method with different thresholding values.

| Thresholding value | RMSE |
|---|---|
| 3 | $3.36 \times 10^{-4}$ |
| 7 | $3.37 \times 10^{-4}$ |
| 10 | $3.31 \times 10^{-4}$ |
| 15 | $3.38 \times 10^{-4}$ |
| 20 | $3.39 \times 10^{-4}$ |

## IV. Conclusion

In this paper, we investigated a method that can detect 3D blood vessel from CT images reconstructed from small number of projection views. The proposed method is used as post-processing step after reconstruction with small number of views. The proposed method is implemented through two stages. First stage is used to reduce noise and low-frequency artifacts. Second stage is based on skeletonization process and converting the skeleton to graph. By evaluating the proposed method on simulation data, we got satisfactory results.

## References


[1] G. T. Herman, *Fundamentals of computerized tomography: image reconstruction from projection*, 2nd ed., Springer-Verlag, 2009

[2] N.Otsu, "A threshold selection method from gray-level histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, pp. 62-66, 1979

[3] T. Lee, R. L. Kashyap and C. Chu, "Building skeleton models via 3-D medial surface/axis thinning algorithms," *CVGIP: Graphical Models and Image Processing*, vol. 56, pp. 462-478, 1994

[4] C.R.F.Maunder, "Algebric Tobology,"*Cambridge Univ. Press*, 1980

[5] J. Lerdsinmongkol, K. Chaisaowong, S. Roongruangsorakarn, T. Kraus and T. Aach, "An Optimal Application of 3D Morphological Operations in the Framework of a Computer-Assisted Diagnosis System,"*The 12th National Computer Science and Engineering Conference*, pp. 659-666,2008

[6] C. Savage, "Depth-first search and the vertex cover problem," *Inform. Process. Lett.*, vol. 14, pp. 233-235, 1982

[7] http://www.imp.uni-erlangen.de/forbild/english/results/index.htm

[8] G. T. Herman and L. B. Meyer, "Algebraic reconstruction techniques can be made computationally efficient," *IEEE Trans. Med. Imaging*, vol. 12, pp. 600-609, 1993